

From Egocentric Experience to Interactive Intelligence: Building the World in First Person

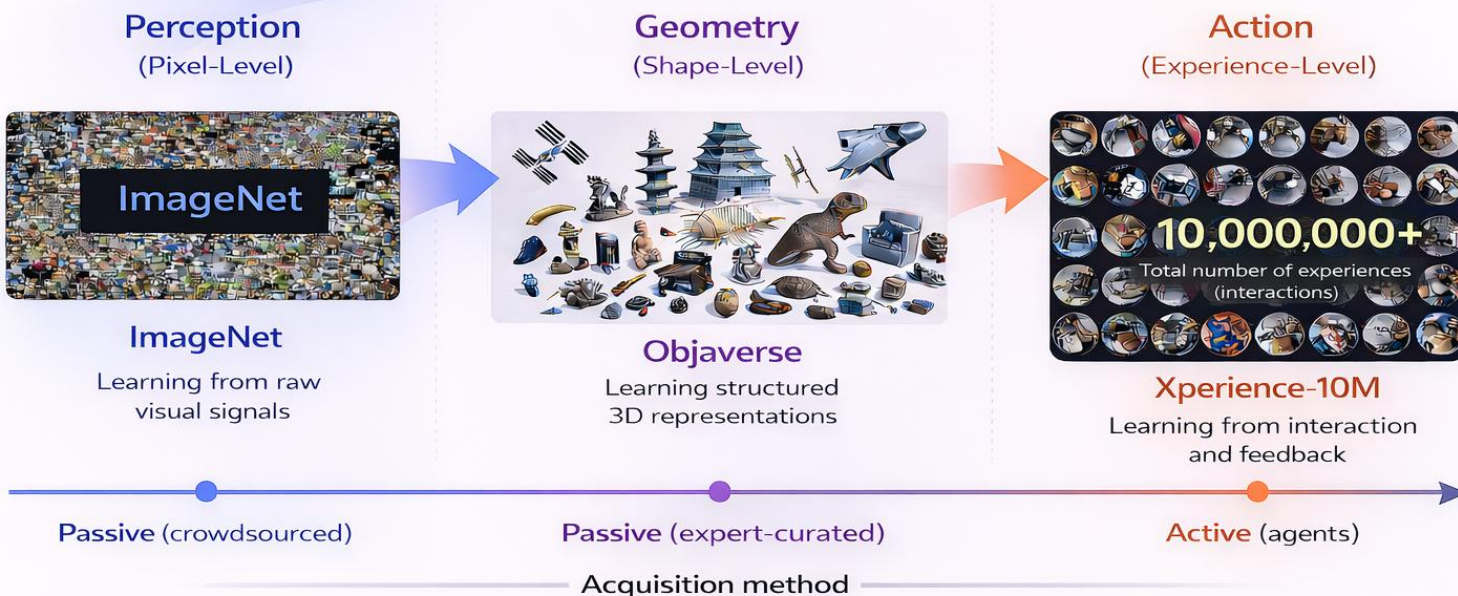
Ziwei Liu

Nanyang Technological University

Three Dimensions of Kant's Laws of Nature



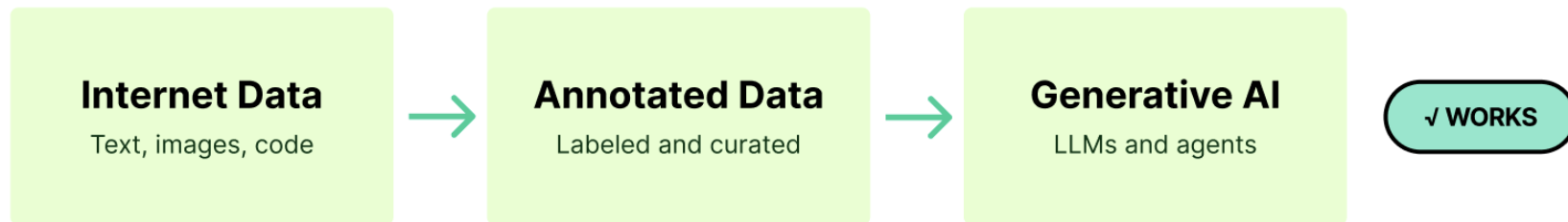
From Perception to Action: The Evolution of Visual Intelligence



The Real-World Data Bottleneck

The same scaling law that powered LLMs is now entering the physical world.

GENERATIVE AI PIPELINE

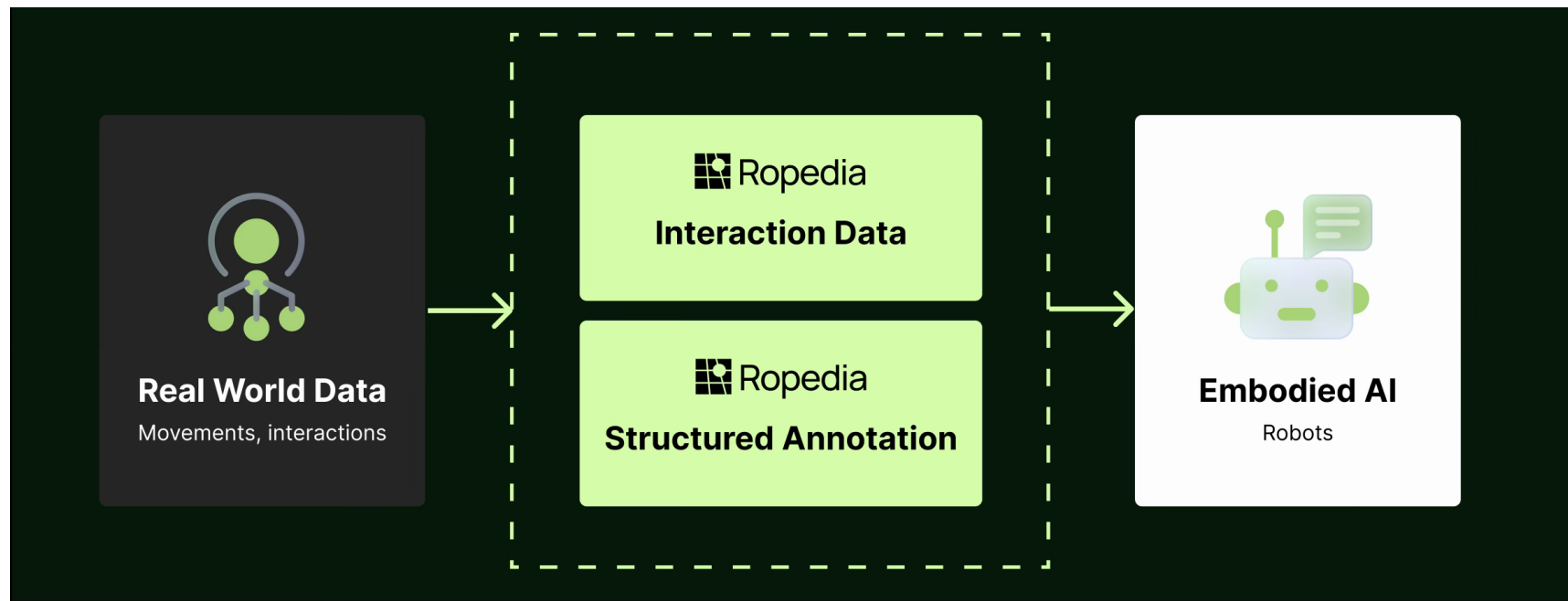


EMBODIED AI PIPELINE



Our Solution

We capture structured real-world interaction at **scale** and **low cost**.



Full-stack Low-cost Portable Data Capture

HOMIE

320g

Super lightweight

360°

Field of View

10x

Faster Deployment

50 μ s

Synchronization

13h

Storage Duration

12.5x

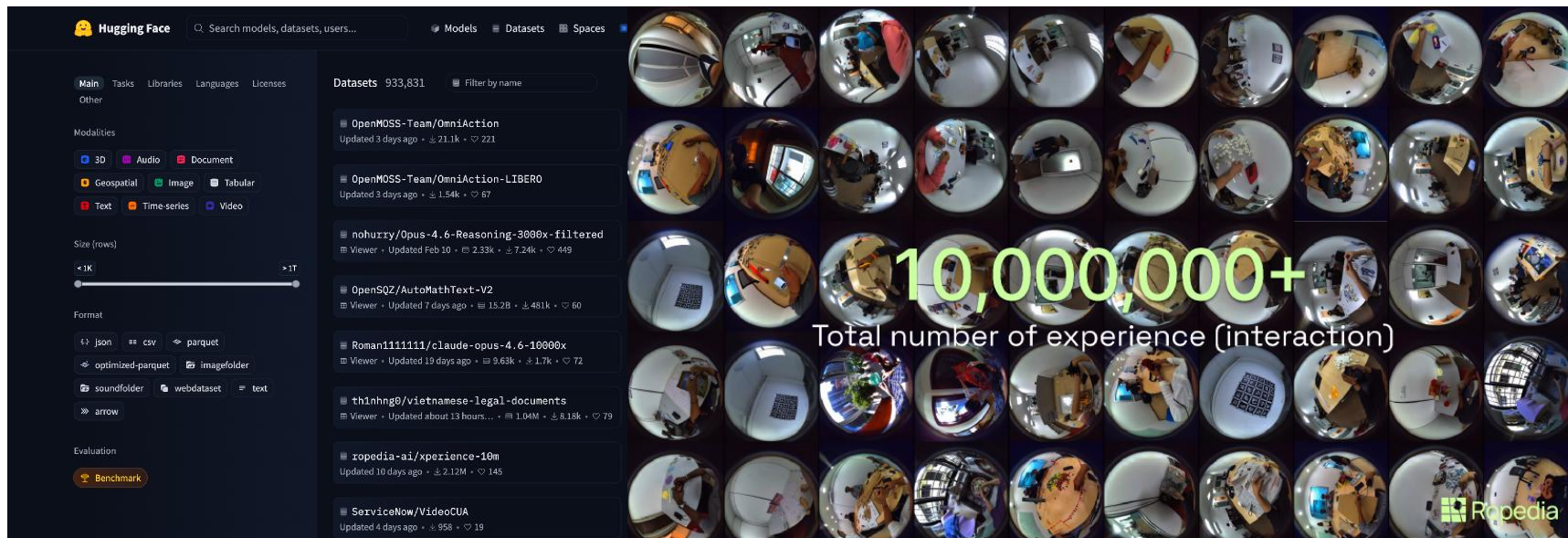
Cost Reduction



Our Dataset: In-the-Wild Human Xperience-10M

2.1M Total Downloads in 1 Week

Top 3 Over ALL Datasets on Hugging Face

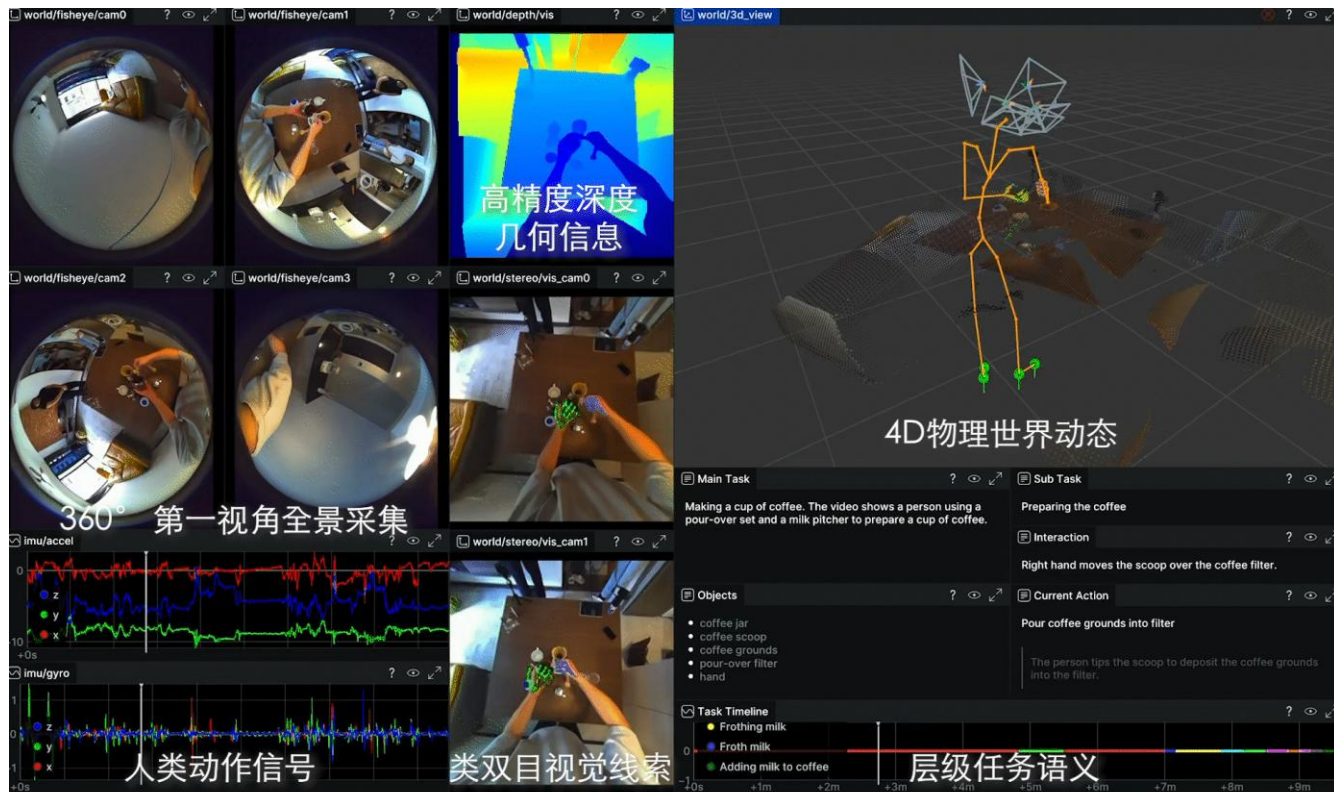


The image shows a screenshot of the Hugging Face datasets page for the dataset 'ropedia-ai/xperience-10m'. The left sidebar displays the Hugging Face interface with search filters for modalities (3D, Audio, Document, Geospatial, Image, Tabular, Text, Time-series, Video) and format (json, csv, parquet, optimized-parquet, imagefolder, soundfolder, webdataset, text, arrow). The main content area lists several datasets, with 'ropedia-ai/xperience-10m' highlighted at the bottom. To the right of the dataset list is a large grid of 100 circular camera view images showing various indoor scenes. Overlaid on this grid is the text '10,000,000+' in large green font, with 'Total number of experience (interaction)' written below it in white. The Ropedia logo is visible in the bottom right corner of the grid.

10,000,000+
Total number of experience (interaction)

ropedia-ai/xperience-10m
Updated 10 days ago • 2.12M • 145

High-Quality Data from Anywhere by Anybody



INTERACTION EPISODES

10M

VIDEO W/O AUDIO

10,000h

RGB FRAME NUMBER

2.88B

MOCAP FRAME

576M

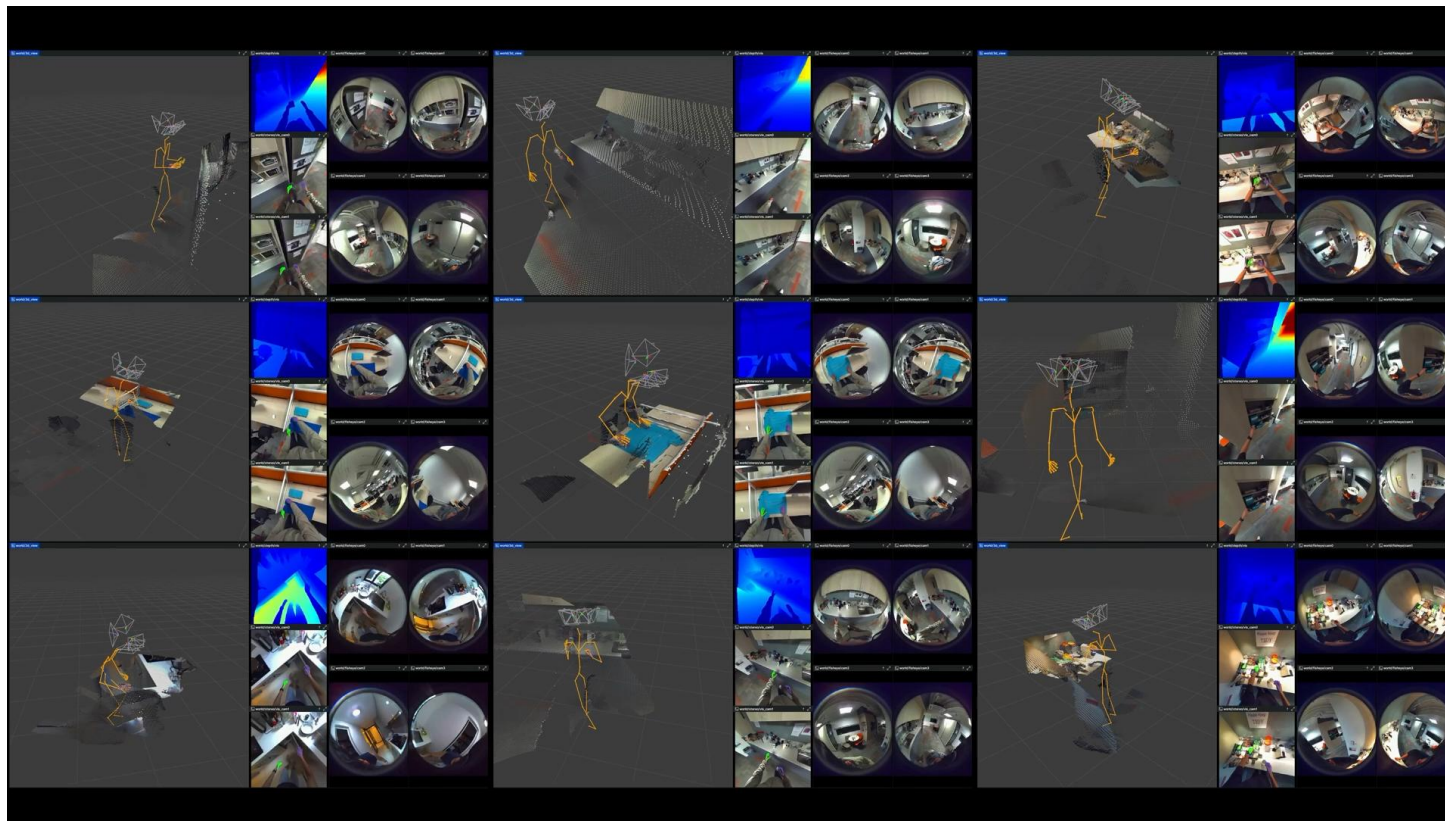
NO. OF OBJECTS

350K

TOTAL STORAGE

~1PB

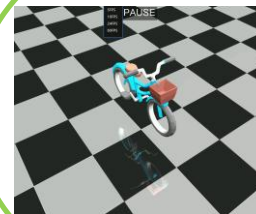
Demo: In-the-wild Data Capture



From **Interactive Experience** to **Simulation-Ready Modeling**



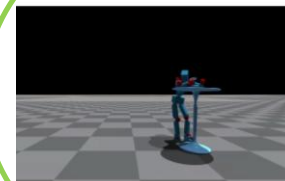
Xperience-10M



Object



Human-Object
Interaction

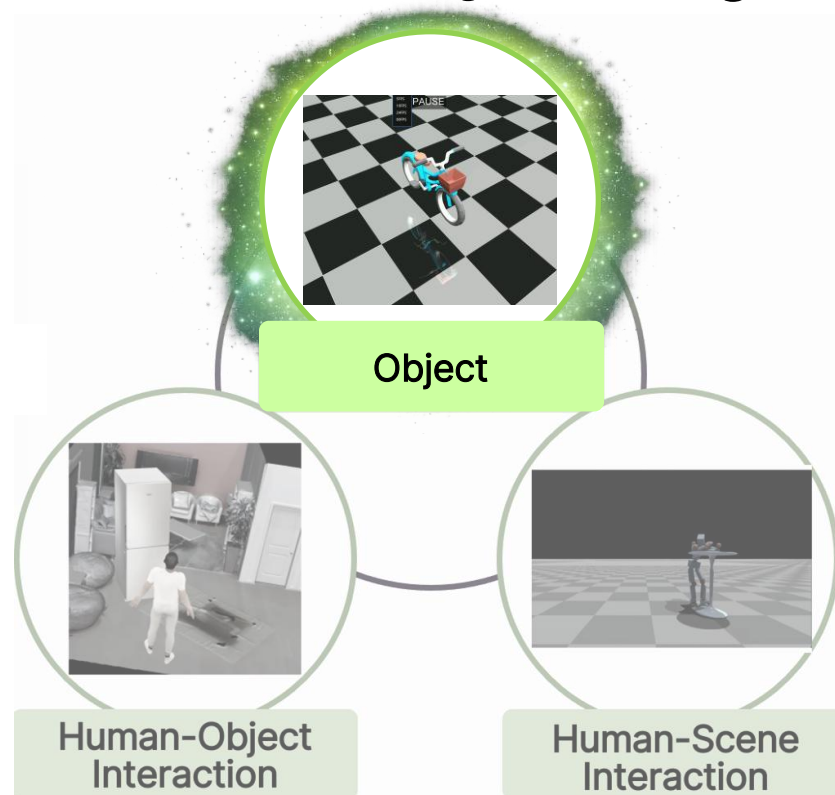


Human-Scene
Interaction

From **Interactive Experience** to **Simulation-Ready** Modeling



Xperience-10M



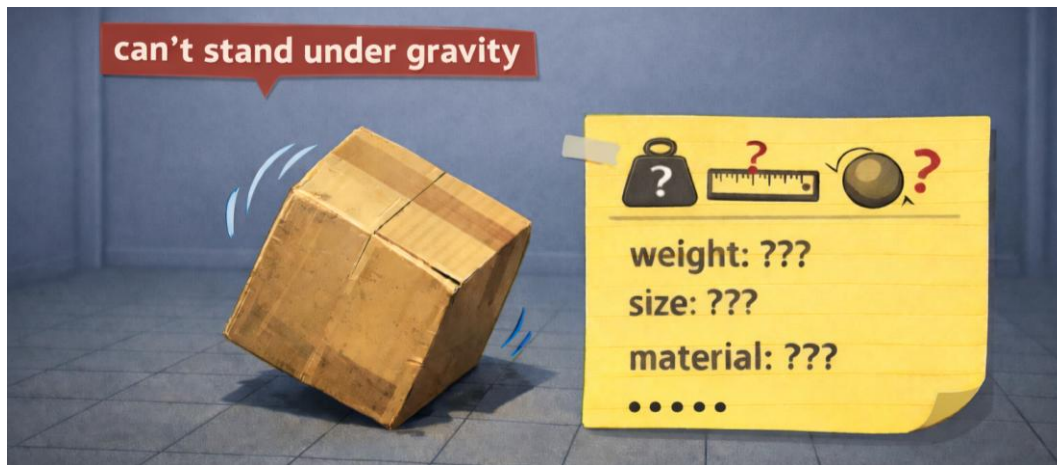
PhysX-Anything: **Simulation-Ready** Physical 3D Assets from Single Image

Ziang Cao, Fangzhou Hong, Zhaoxi Chen, Liang Pan, Ziwei Liu.

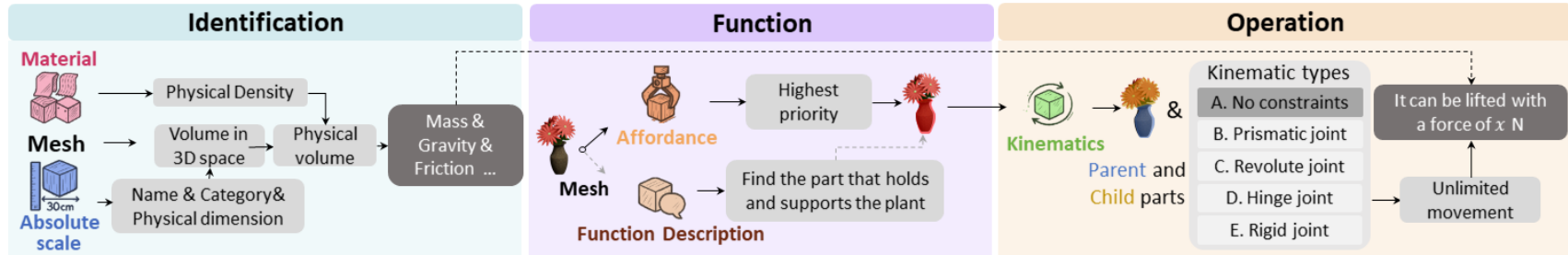


Challenge

- ▶ Existing 3D generation primarily emphasizes geometries and textures while **neglecting physical-grounded modeling**, hampering their real-world application in physical domains like simulation and embodied AI.

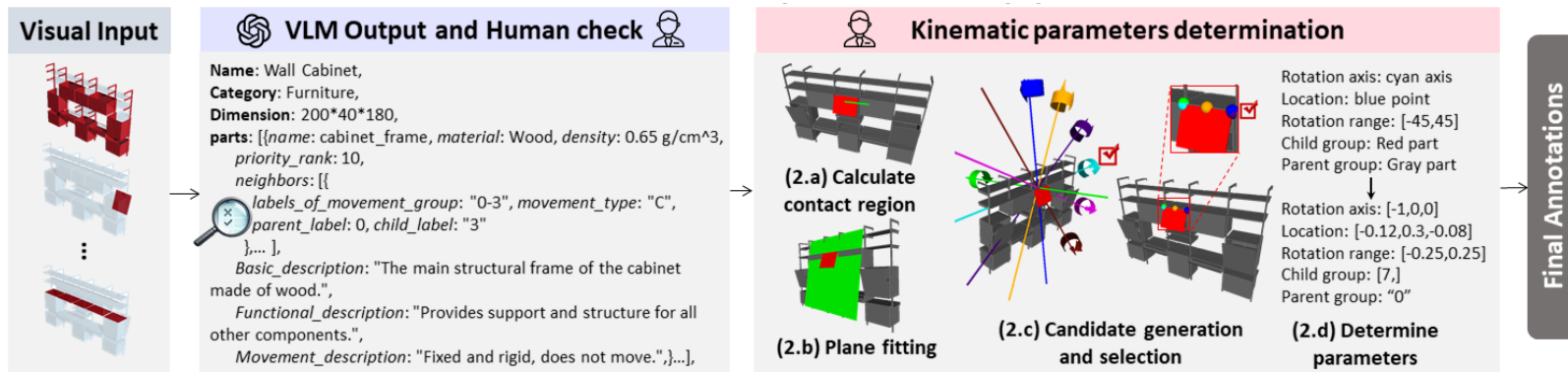


Definition of Physical Properties



- **Identification** – Determining the basic nature of the object
Absolute scaling and material (Young's modulus, Poisson's ratio, and density)
- **Function** – Understanding its potential applications
Functional affordance analysis and function descriptions
- **Operation** – Detailed usage methodologies
Kinematic Parameter

Human-in-the-loop Annotation Pipeline



► Preliminary Data Acquisition

GPT-4o



Human Check

► Kinematic Parameter Determination

Contact Region



Plane Fitting

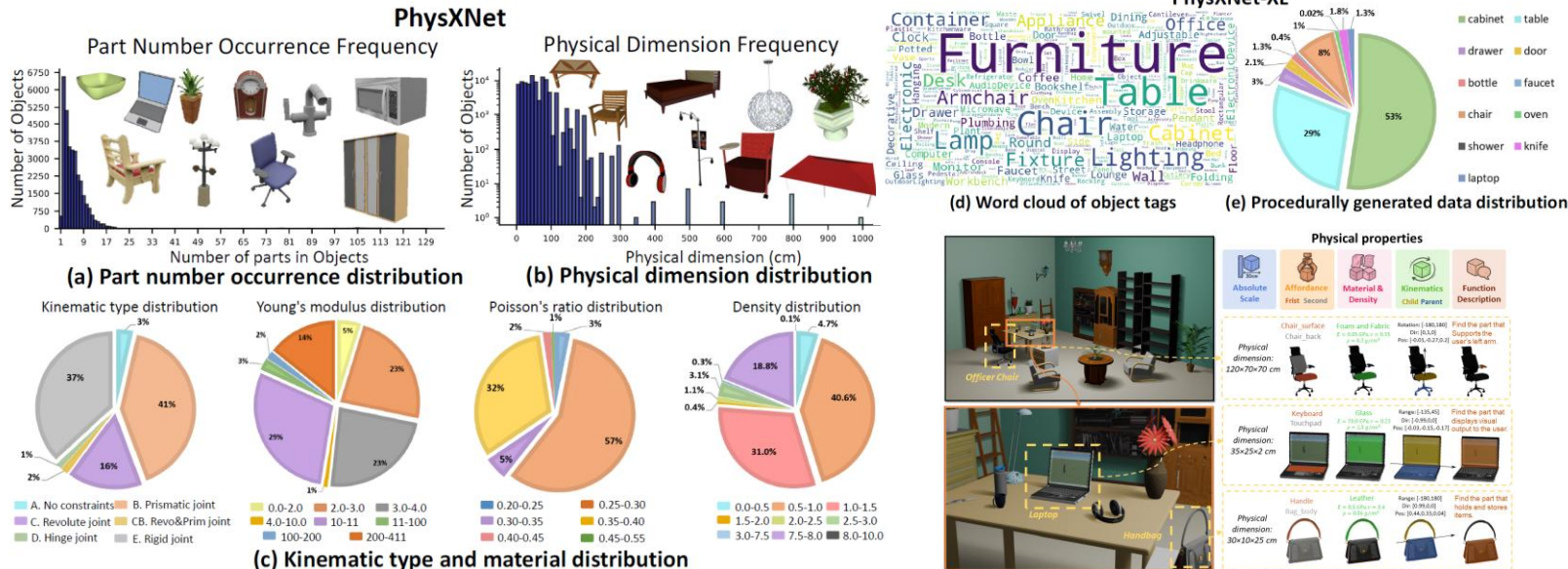


Candidate Generation
and Selection



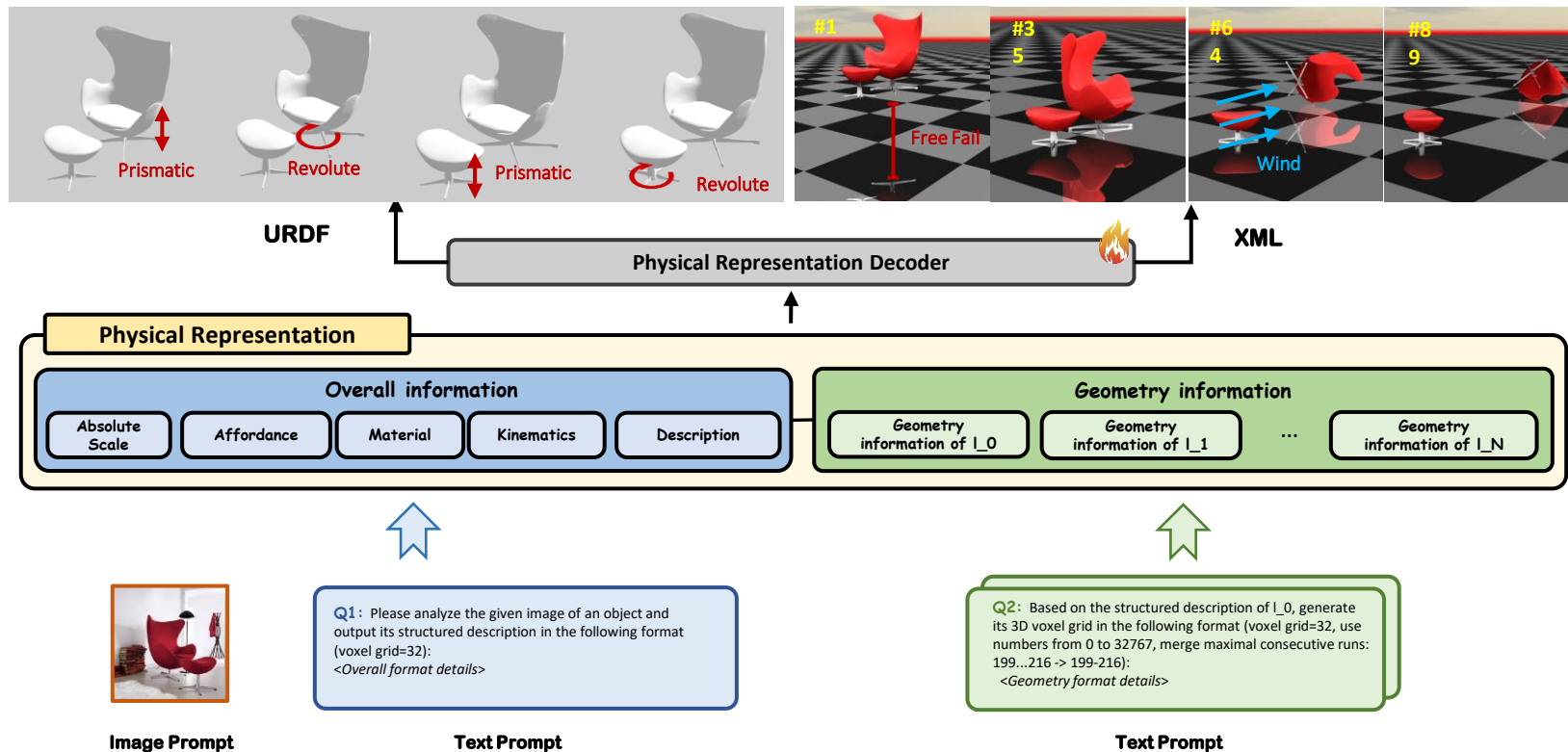
Kinematic Parameter
Determination

PhysXNet & PhysXNet-XL

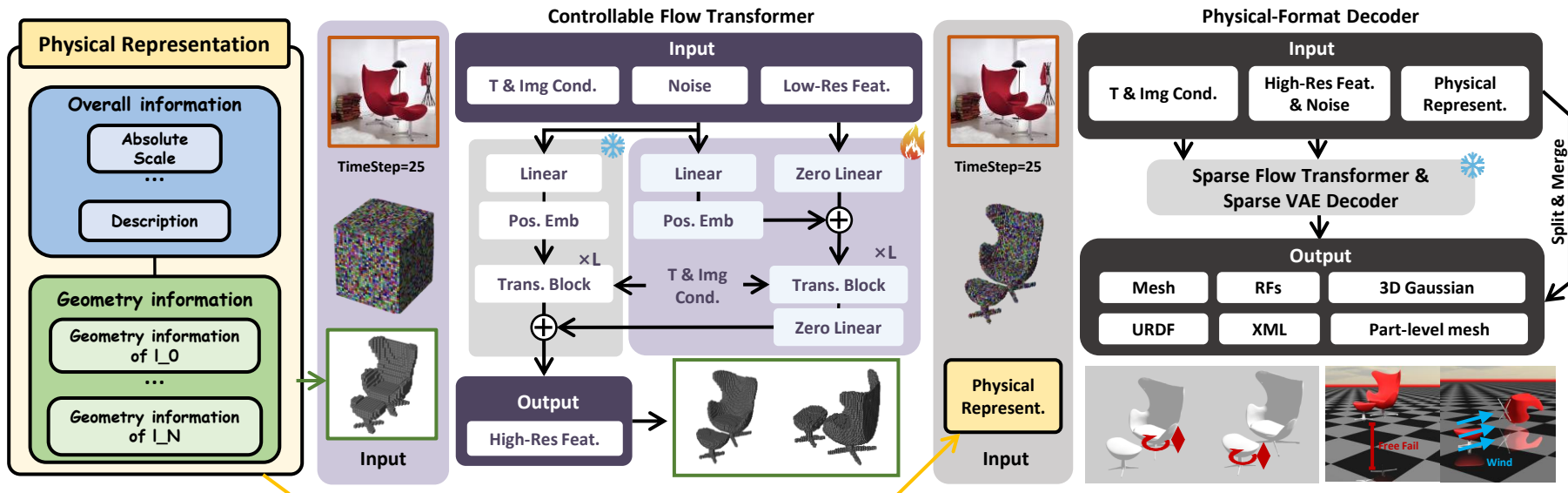


We propose **PhysXNet** and **PhysXNet-XL** – the first comprehensive physical 3D dataset with over **26K** and **6M** richly annotated 3D objects

Key Idea: Interactive 3D Object as **Physical Code**



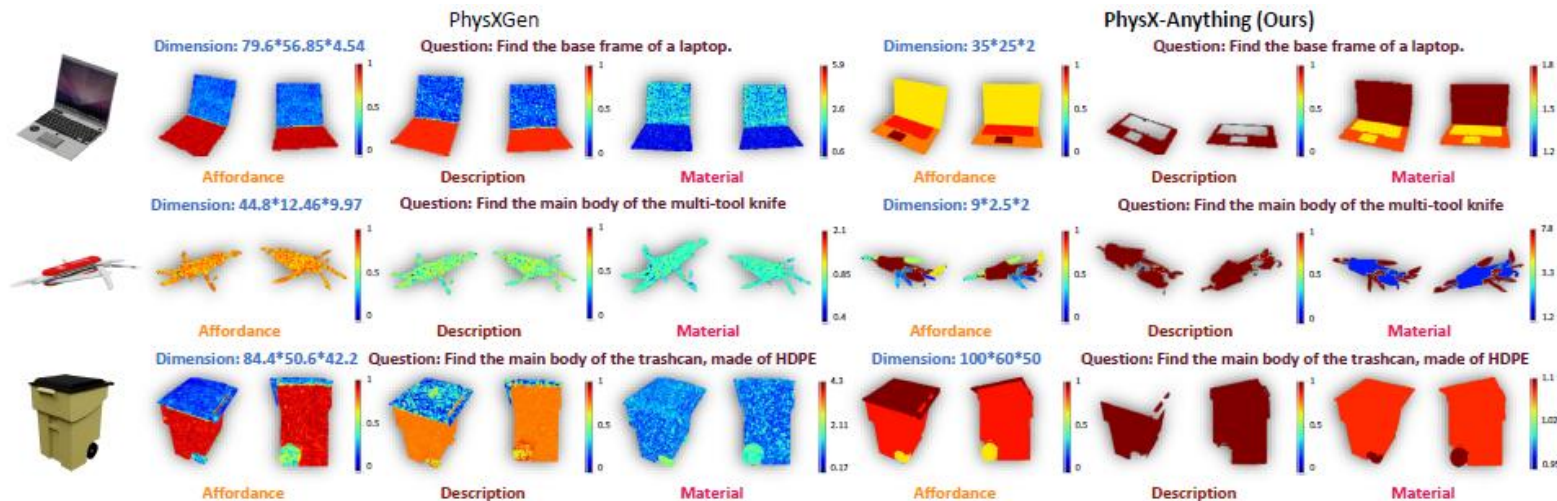
Method: Physical Representation Decoder



Results: Qualitative Comparisons



Results: Qualitative Comparisons



Results: Quantitative Comparisons

Methods	Geometry			Physical Attributes				
	PSNR $\uparrow$	CD $\downarrow$	F-score $\uparrow$	Absolute scale $\downarrow$	Material $\uparrow$	Affordance $\uparrow$	Kinematic parameters (VLM) $\uparrow$	Description $\uparrow$
URDFormer [11]	7.97	48.44	43.81	—	—	—	0.31	—
Articulate-Anything [16]	16.90	17.01	67.35	—	—	—	0.65	—
PhysXGen [3]	20.33	14.55	76.3	43.44	6.29	9.75	0.71	12.89
PhysX-Anything (Ours)	20.35	14.43	77.50	0.30	17.52	14.28	0.83	19.36

Results: Simulated Objects

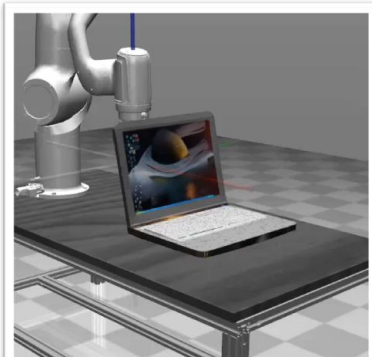


Results: Simulated Objects

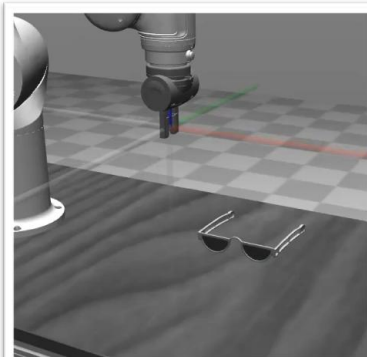


Results: Simulated Interactions

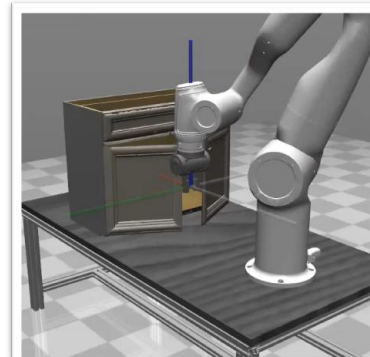
Laptop Closing



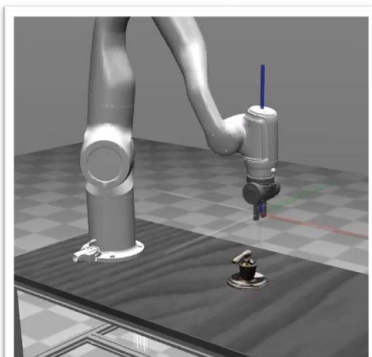
Eyeglass Temple Folding



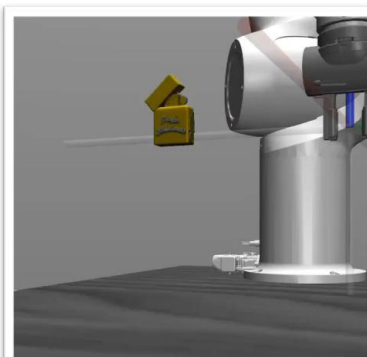
Door Opening and Closing



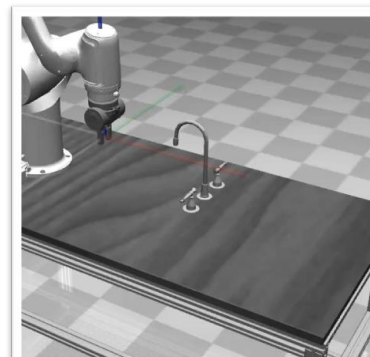
Handle Manipulation



Lighter Snapping Open



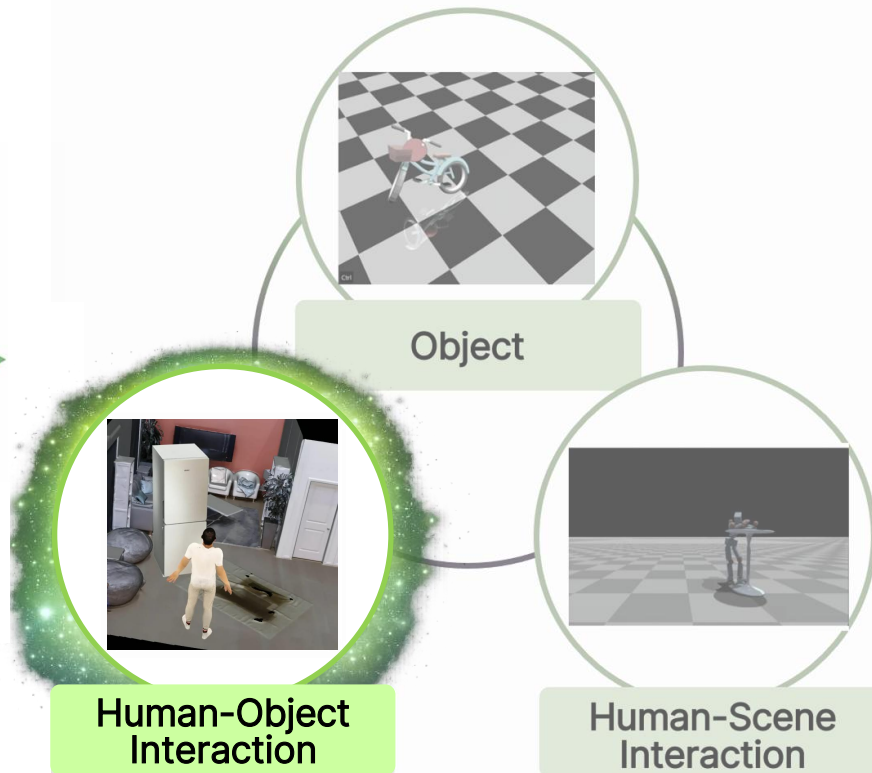
Faucet Switch Manipulation



From **Interactive Experience** to **Simulation-Ready** Modeling



Xperience-10M



ArtHOI: Articulated Human-Object Interaction Synthesis by 4D Reconstruction from Video Priors

Zihao Huang, Tianqi Liu, Zhaoxi Chen, Shaocong Xu, Saining Zhang,
Lixing Xiao, Zhiguo Cao, Wei Li, Hao Zhao, Ziwei Liu

RGB Rendering

✗ TRUMANS
✗ ZeroHSI
✓ Ours

Articulation

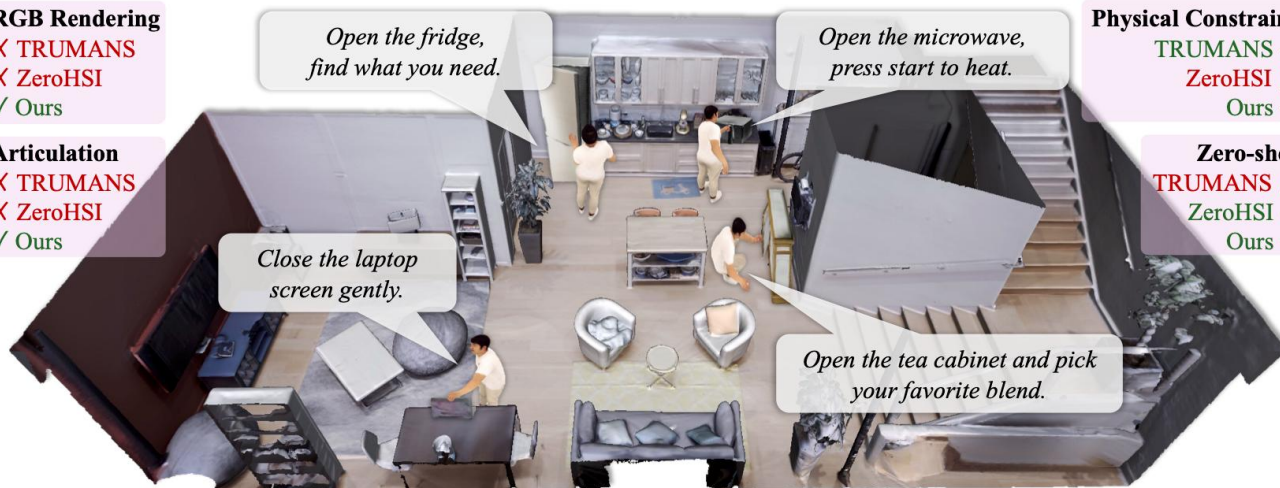
✗ TRUMANS
✗ ZeroHSI
✓ Ours

Physical Constraint

TRUMANS ✓
ZeroHSI ✗
Ours ✓

Zero-shot

TRUMANS ✗
ZeroHSI ✓
Ours ✓



Task: Reconstruction of Articulated HOI

Input



3D Assets + Text Prompt

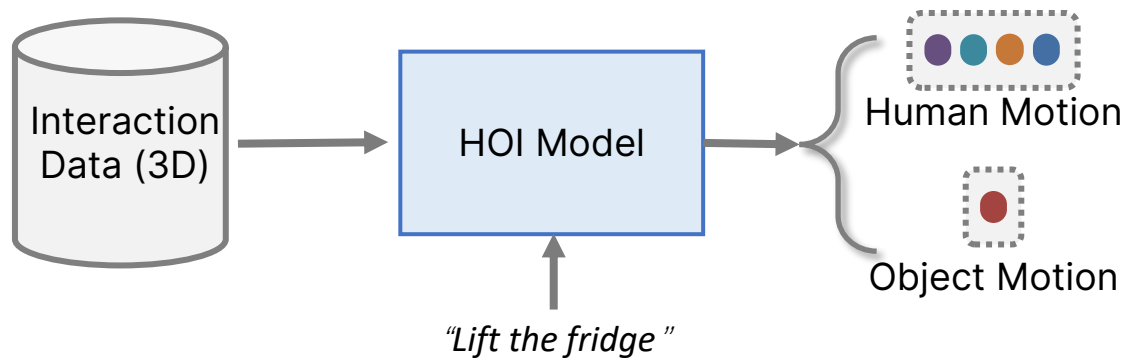
Reconstruction

Output

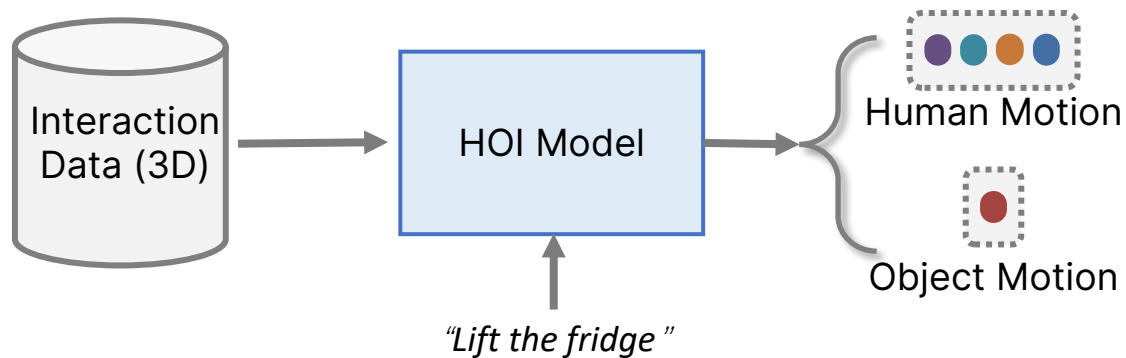


Articulated Human-object Interaction

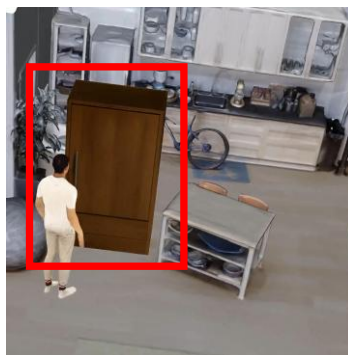
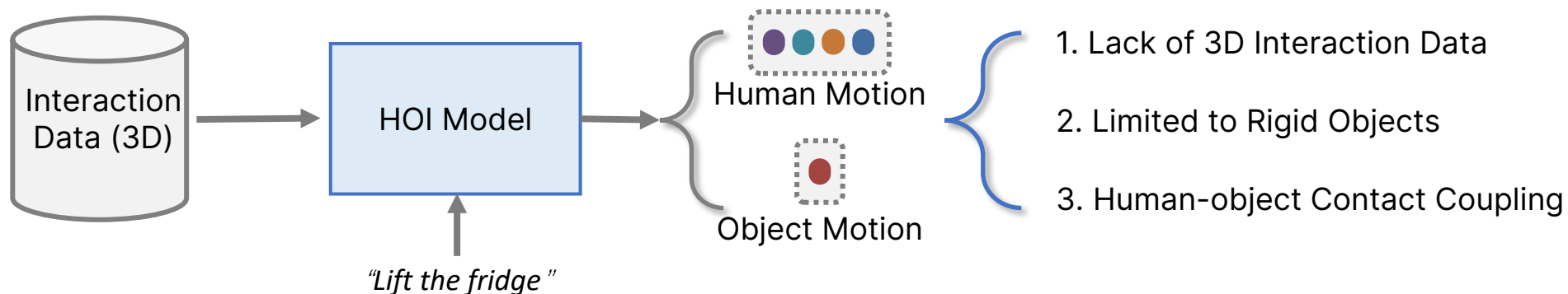
Challenge



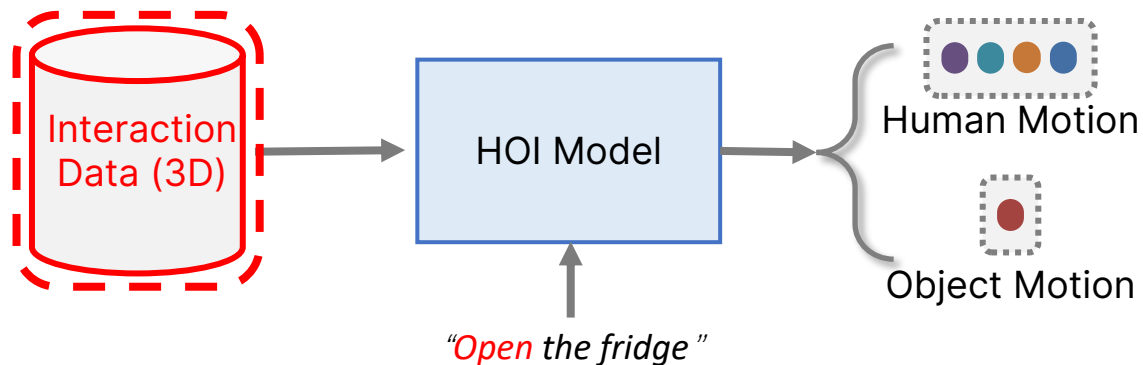
Challenge



Challenge

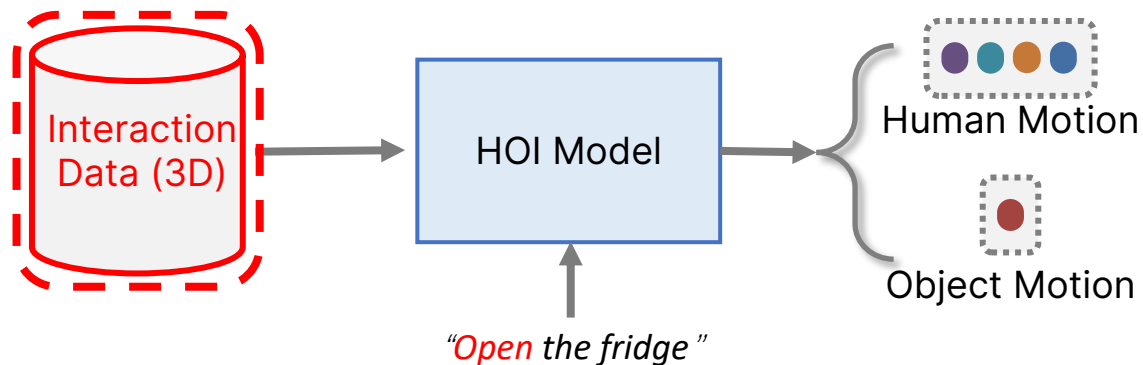


Key Idea



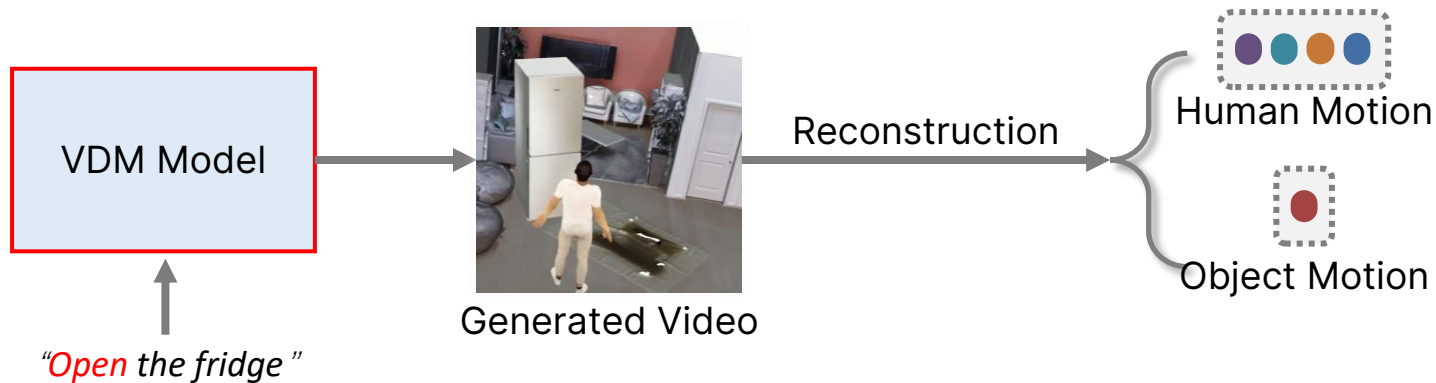
- 1. Lack of 3D Interaction Data
- 2. Limited to Rigid Objects
- 3. Human-object Contact Coupling

Key Idea



- 1. Lack of 3D Interaction Data
- 2. Limited to Rigid Objects
- 3. Human-object Contact Coupling

Key Idea



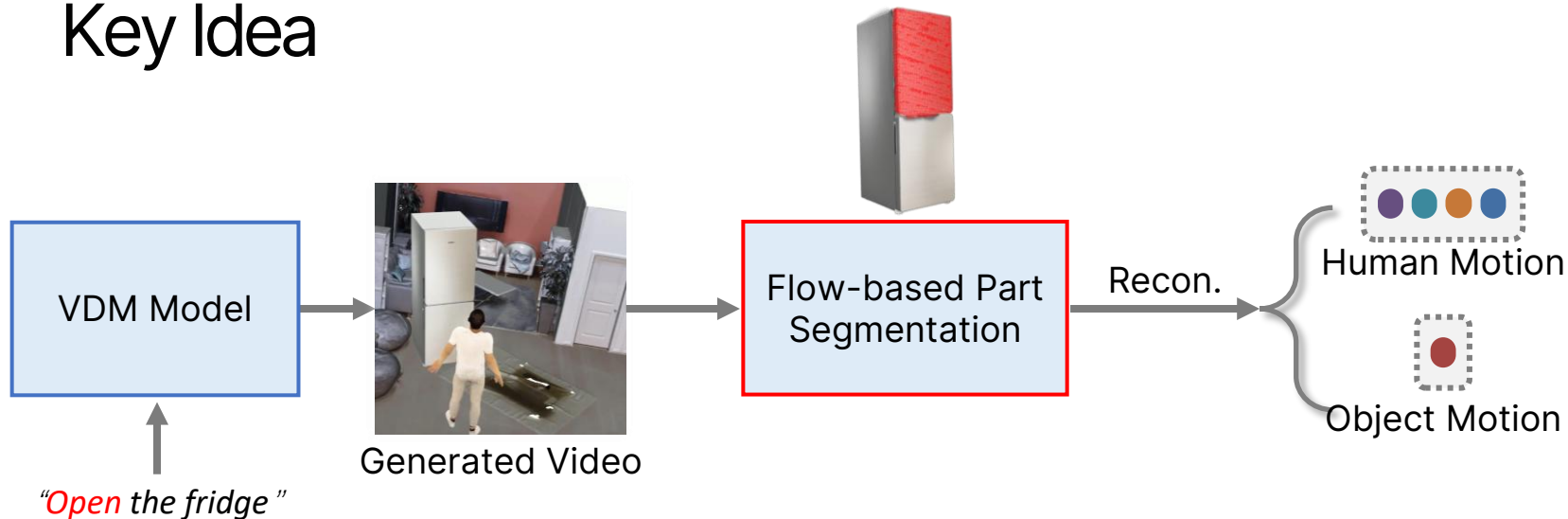
~~1. Lack of 3D Interaction Data~~

Video Prior

2. Limited to Rigid Objects

3. Human-object Contact Coupling

Key Idea



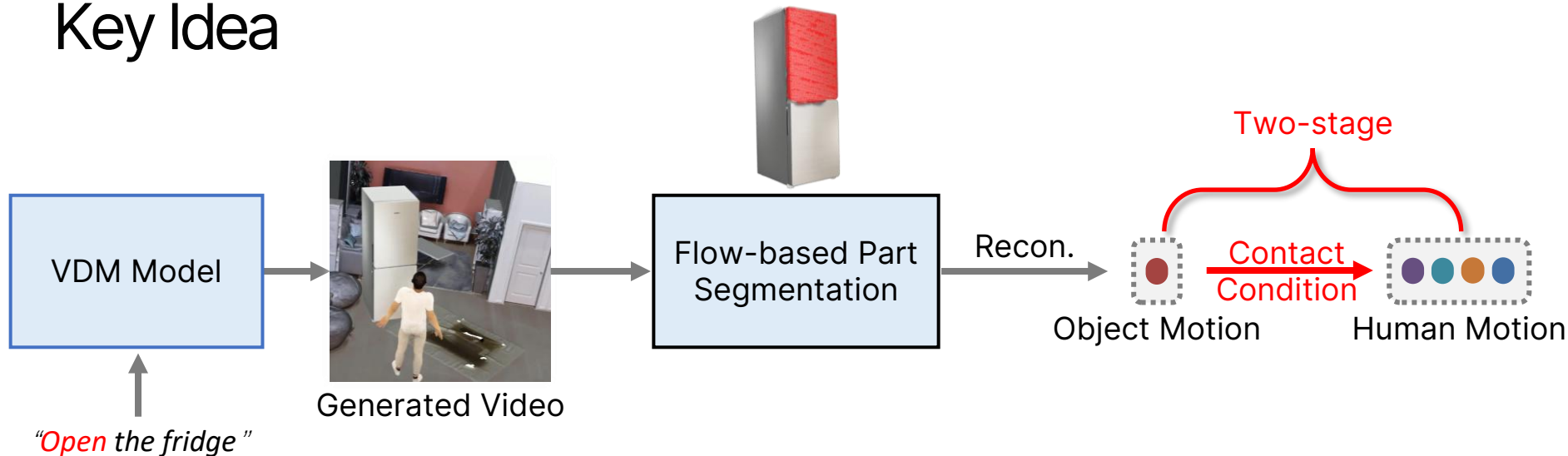
~~1. Lack of 3D Interaction Data~~

~~2. Limited to Rigid Objects~~

3. Human-object Contact Coupling

Flow-based Part Segmentation

Key Idea



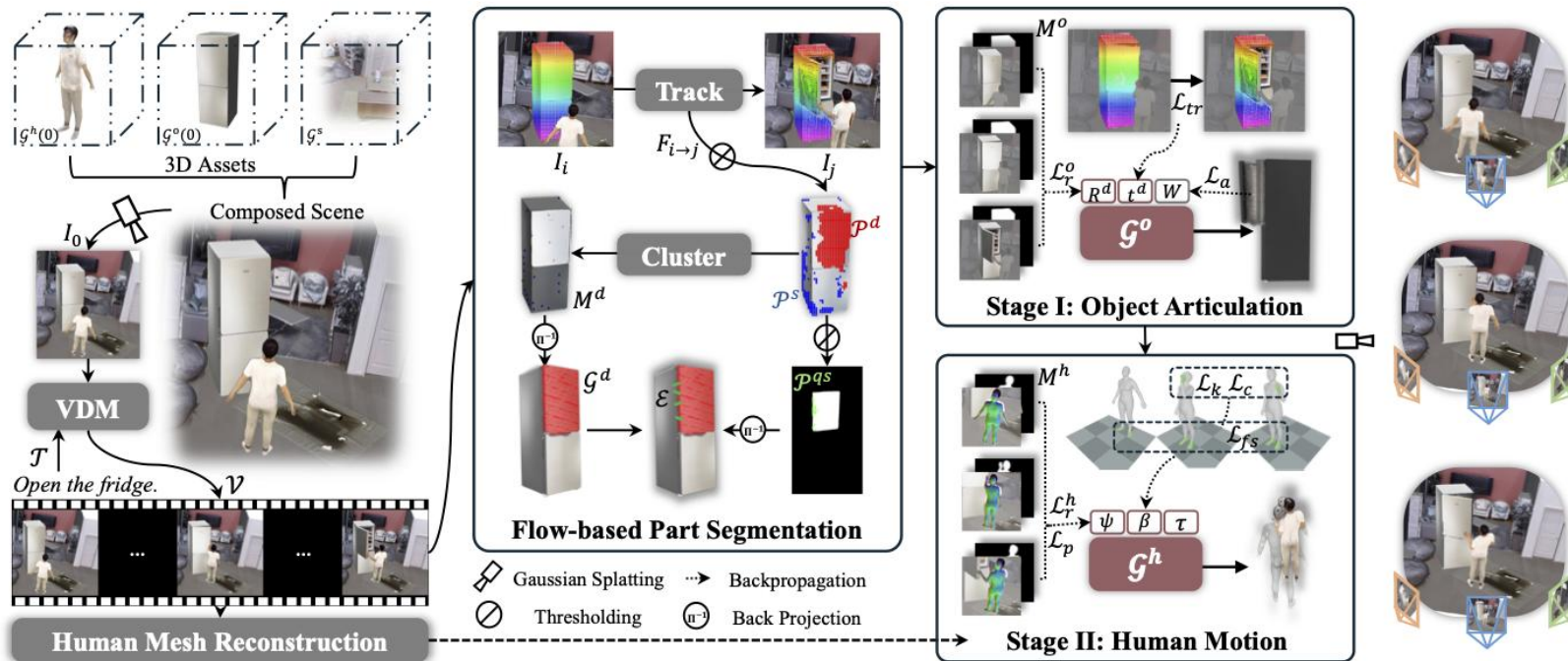
1. ~~Lack of 3D Interaction Data~~

2. ~~Limited to Rigid Objects~~

3. Human-object Contact Coupling

Two-stage Decoupled Reconstruction

Method

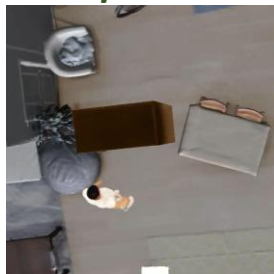


Results: Qualitative Comparisons

Front View

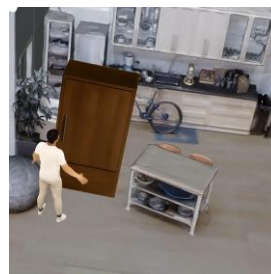


Top View

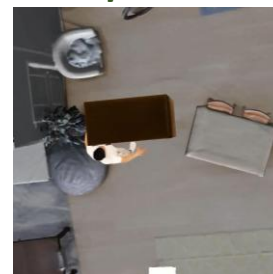


CHOIS

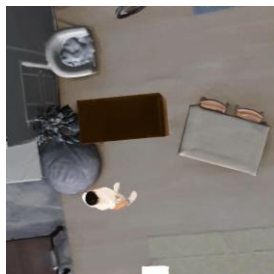
Front View



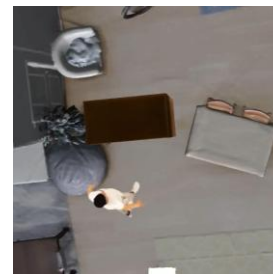
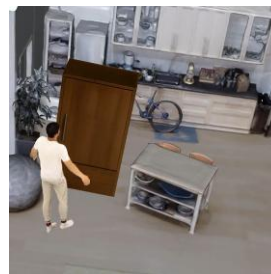
Top View



ZeroHSI



LINGO



Ours

“Open the cabinet, look inside.”

Results: Qualitative Comparisons

Front View



Top View

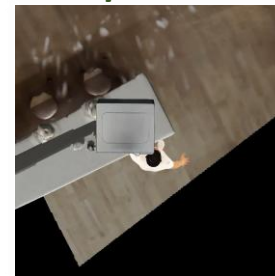


CHOIS

Front View



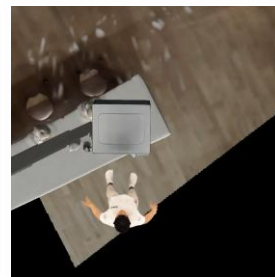
Top View



ZeroHSI



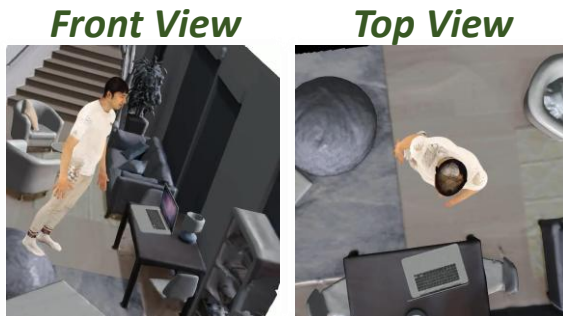
LINGO



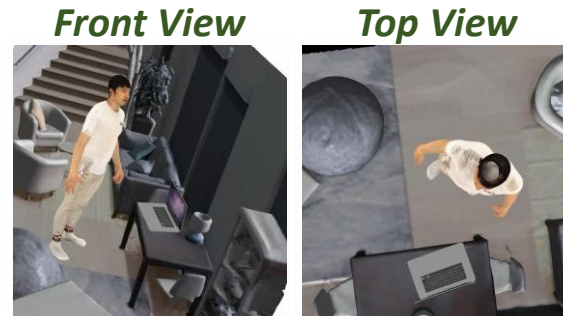
Ours

“Open the microwave”

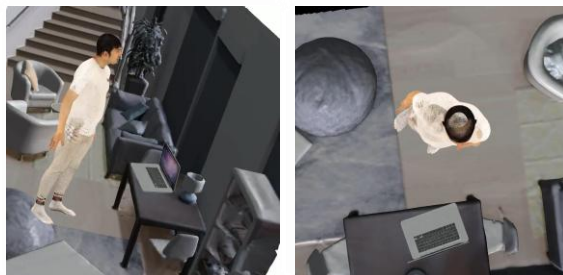
Results: Qualitative Comparisons



CHOIS



ZeroHSI



LINGO



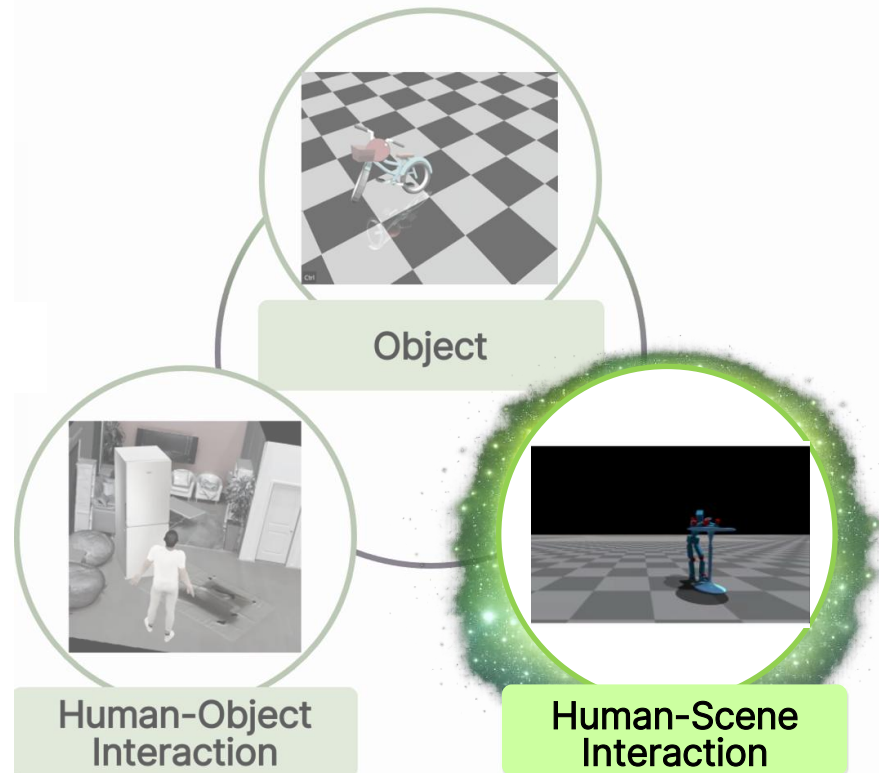
Ours

“Close the laptop screen gently.”

From **Interactive Experience** to **Simulation-Ready** Modeling

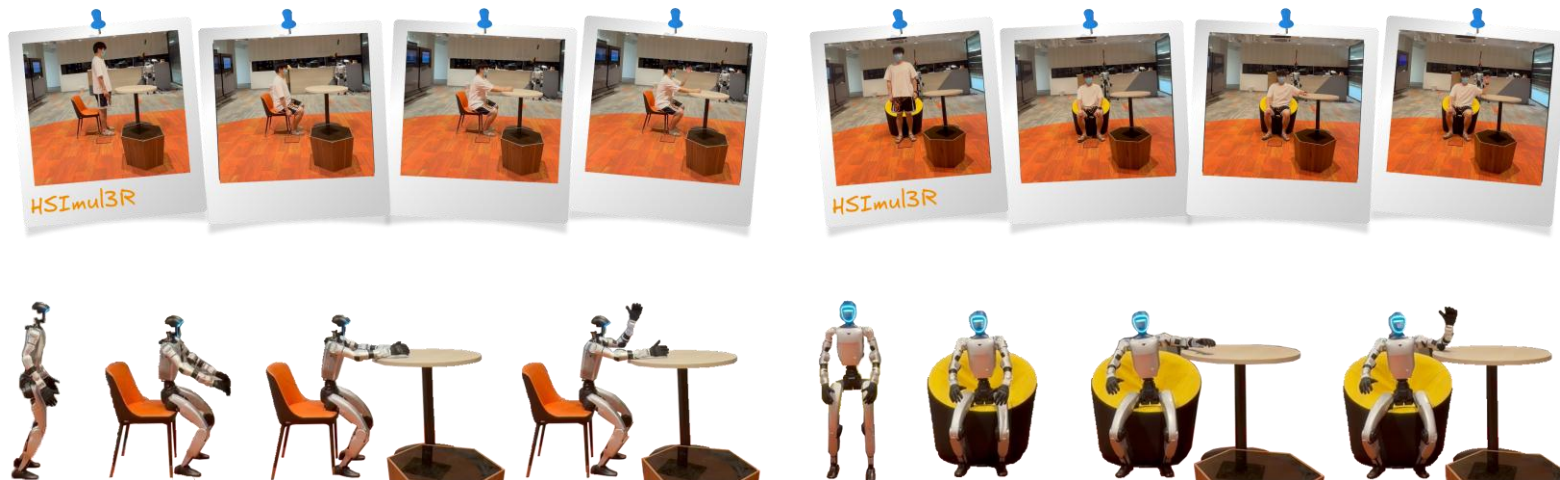


Xperience-10M



HSImul3R: Physics-in-the-Loop Reconstruction of **Simulation-Ready** Human-Scene Interactions

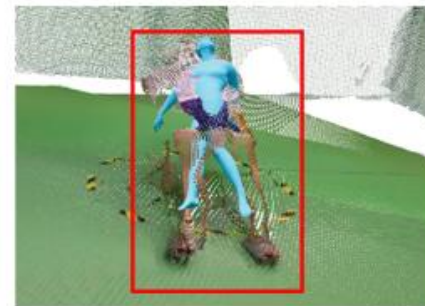
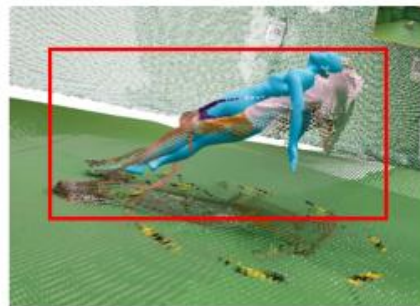
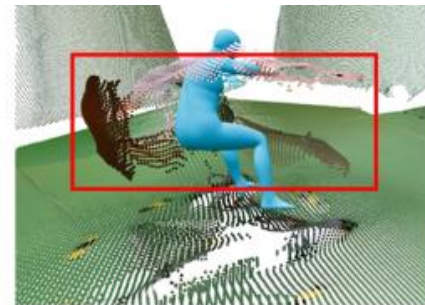
Yukang Cao, Haozhe Xie, Fangzhou Hong, Long Zhuo, Zhaoxi Chen, Liang Pan, Ziwei Liu



Challenge

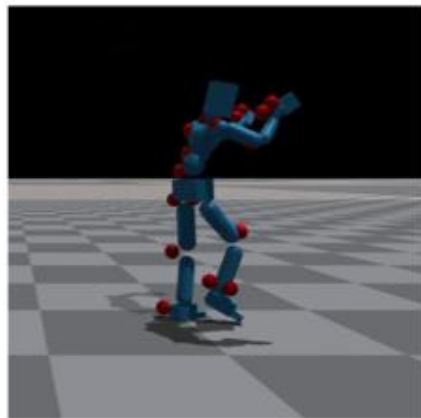
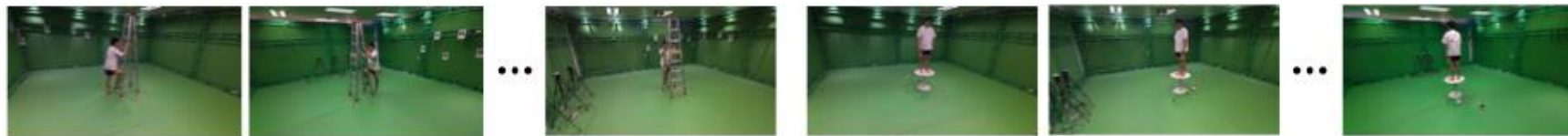


Inputs

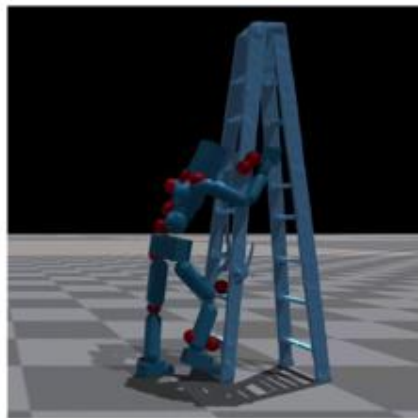


HSfM

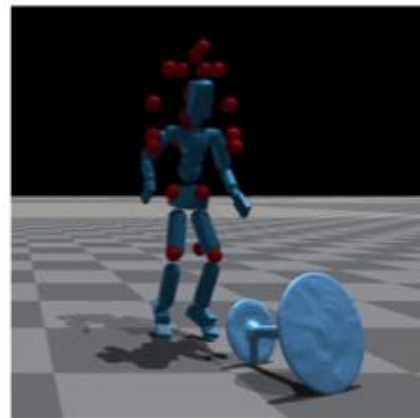
Challenge



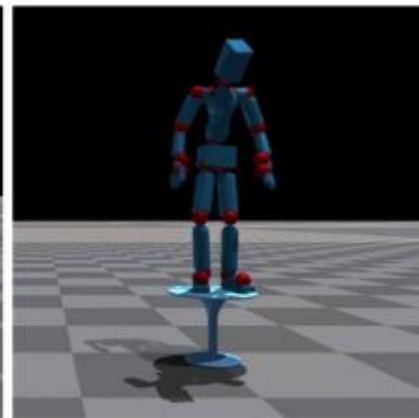
HSfM



Ours



HSfM



Ours

Key Idea: Leverage **Simulator Interaction Feedback** to Optimize Human Motion and Scene Geometry



Casual Capture

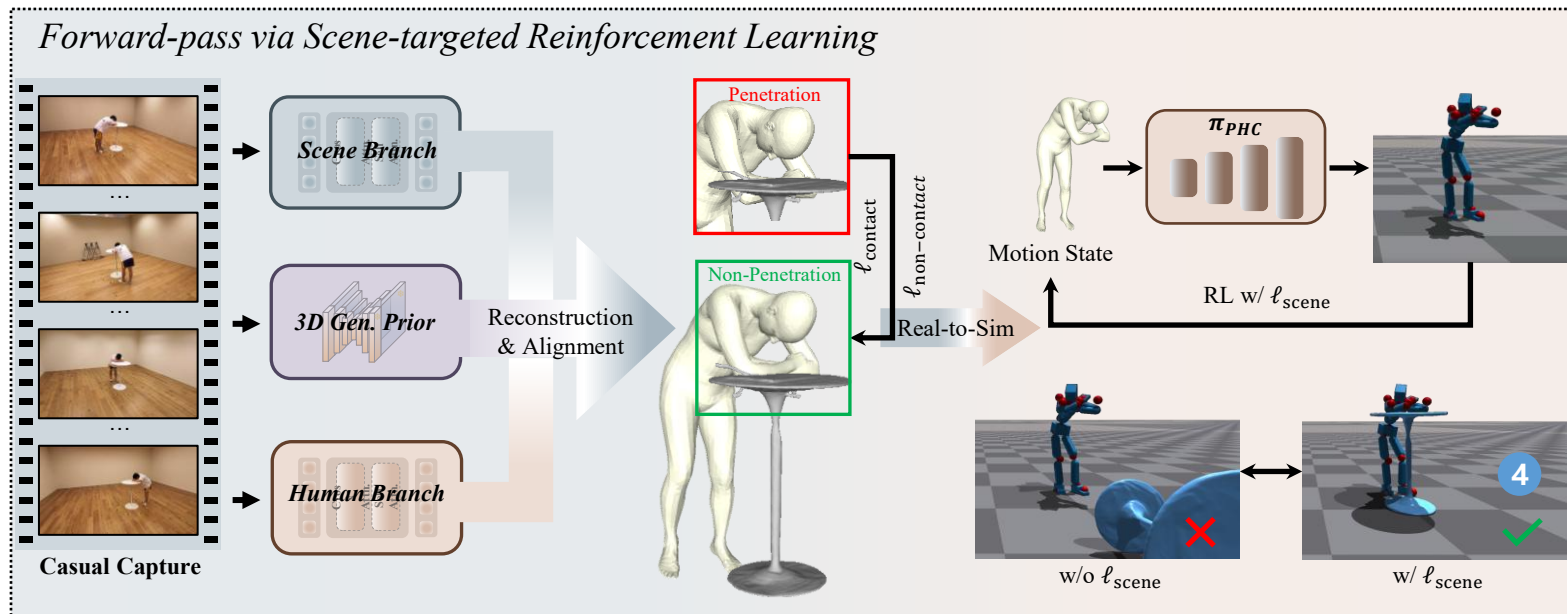


Optimize Human Motion



Optimize Scene Geometry

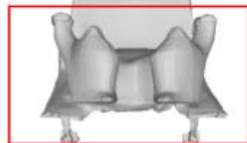
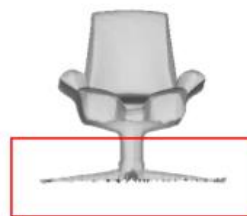
Stage I: Motion-Optimization via Scene-targeted Reinforcement Learning



Issue: Structural Suboptimality in the Reconstructed Scene



Inputs



DSO*

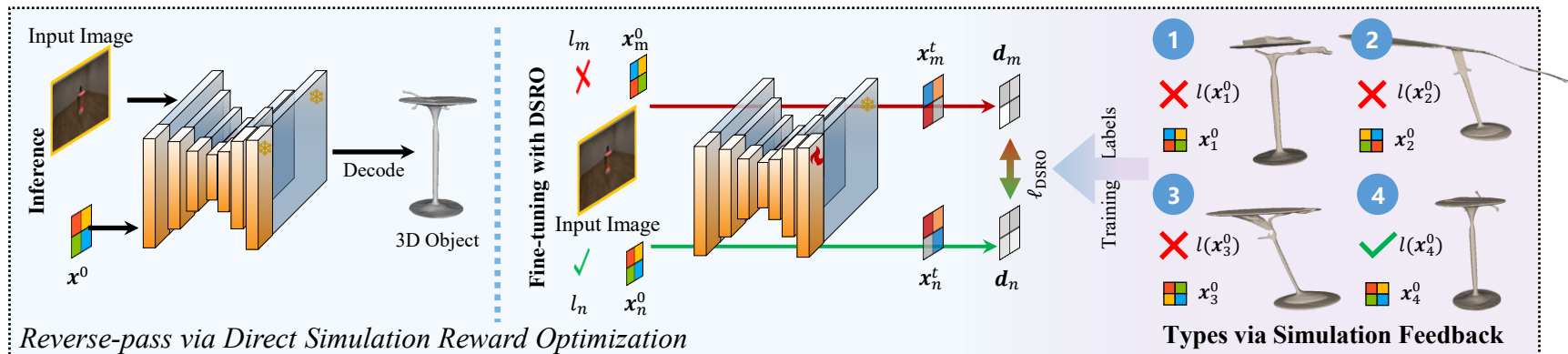


MIDI



Ours

Stage II: Scene-Optimization via Simulator Feedback



Dataset: HSI Bench

- a new dataset comprising 19 objects and over 50 motion sequences recorded by two male and one female participants, totaling 300 unique interaction instances.



Dataset: HSiBench



Dataset: HSiBench



Results: Quantitative Comparisons

Table 1: Quantitative comparison regarding human-scene-interaction reconstruction and simulation. We assess (1) post-simulation interaction stability, (2) human-scene penetration in the 3D reconstruction, and (3) changes in the human motions after simulation. Our method consistently outperforms the baseline method (HSfM) and different variants.

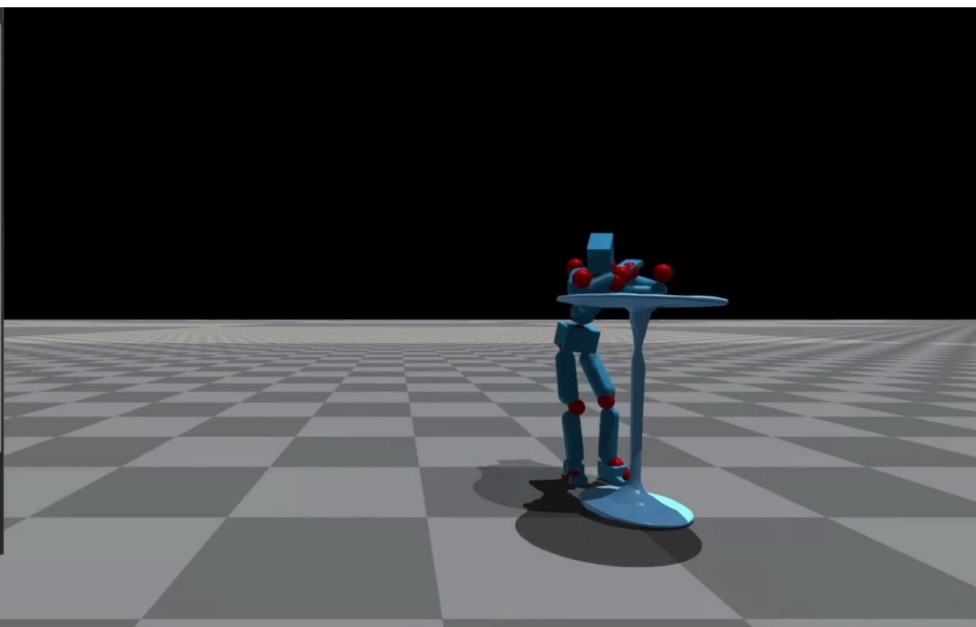
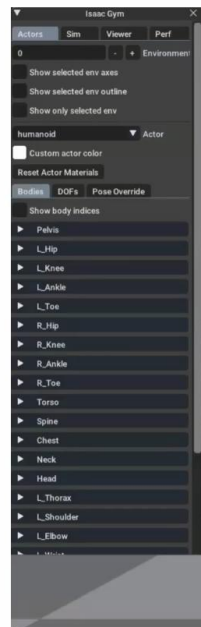
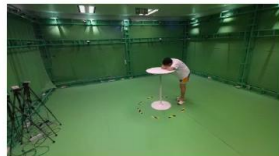
Method	Stability-HSI (%) $\uparrow$			Scene Penetration	Human Motion Quality	
	Easy	Medium	Hard	SP-3D (%) $\downarrow$	W-MPJPE $\downarrow$	PA-MPJPE $\downarrow$
HSfM [52]	10.52	4.50	2.66	69.51	5.02	2.79
V1	13.96	8.81	4.17	77.12	6.18	3.20
V2	39.56	22.71	7.05	-	4.91	2.71
V3	42.57	23.84	10.18	-	4.60	2.42
V4	29.56	16.62	5.17	-	4.57	2.39
Ours	53.68	30.56	13.92	22.9	4.09	2.17

Results: Quantitative Comparisons

Table 2: Quantitative comparison regarding image-to-3D generation quality. Stability-HSI’ denotes the ratio of simulations in which the human–scene interaction remains stable, while “Stability-Gravity”(SG) refers to placing the object alone in the simulator and evaluating whether it can stand stably under gravity. Note that we fine-tune DSO on pre-trained MIDI model for fair comparison.

Method	Stability-HSI (%) $\uparrow$			SG (%) $\uparrow$	CD $\downarrow$	F-Score $\uparrow$
	Easy	Medium	Hard			
MIDI [23]	29.56	16.62	5.17	79.19	0.198	81.95
DSO* [31]	38.75	25.91	7.88	87.23	0.191	86.26
Ours	53.68	30.56	13.92	91.50	0.173	88.25

Results: Real-to-Sim



Results: Sim-to-Real



We are moving from a world that is seen,
to a world that is **experienced**

